

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2013

Subject: 394171 Mathematics I

Section: 15-16

Date: 29 July 2013

Time: 13.00-15.00

Name: _____ ID: _____ Class: _____

Instructions

1. Cheating will result in failure of all classes registered for the current semester.
Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Textbooks and dictionary are NOT allowed.
4. Calculators are allowed.
5. Electronic communication devices are NOT allowed in the examination room.
6. The examination has 7 pages (including this page), 11 questions and please do only 9 questions and a total score of 90 marks.
7. Write all your answers on this examination sheet.

Question 1. Given $U = \{1, 2, 3, 4, \dots, 9\}$, $A = \{2, 3, 5, 7\}$, and $B = \{2, 4, 6, 8\}$. Find the set $A - B$ and the number of element of $P(A \cap B')$. (10 marks)

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Question 2. . Given $A \cap B = \{-1, 0, 2, 3\}$, $A \cap C = \{-1, 0\}$, $B \cap C = \{-1, 0, 4\}$, $A \cup B = \{-1, 0, 2, 3, 4, 5\}$, and $A \cup C = \{-1, 0, 2, 3, 4\}$. Find the set $(B - A) \cap C$. (10 marks)

Question 3. In a family survey asking 1,000 households about their household applications, it was found that each household had at least one of the following applications, refrigerator, iron and stove. It was also found that

240 households do not have a refrigerator,

680 households do not have an iron,

530 households do not have a stove,

120 households have a refrigerator and an iron,

330 households have a refrigerator and a stove,

150 households have an iron and a stove.

How many households have only a refrigerator? (10 marks)

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Question 4. Find the set of all solutions of the inequality $x^2 + 6x \geq -2$. (10 marks)

Question 5. Solving the inequality $1 \leq \frac{2x-4}{3-x} \leq 2$. (10 marks)

Question 6 Find an identity of the operator, $a * b = \frac{a+3b}{3}$, where a, b are real numbers. (10 marks)

Question 7 Given a, b, c and d are real number where $b, d, \neq 0$. To prove that

$$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd} \quad (10 \text{ marks})$$

Question 8 Using trigonometric identities to transform the left side of the equation

$$\frac{\tan \beta + \cot \beta}{\tan \beta} = \operatorname{cosec}^2 \beta$$

into the right side. (10 marks)

Question 9 Using a unit trigonometric circle to evaluate the expression

$$\frac{\cos^2 \frac{\pi}{6} - \cos^2 \frac{\pi}{3} - \cos^2 \frac{3\pi}{2}}{\cos^2 \frac{\pi}{3} - \cos^2 \frac{\pi}{6} - \sin^2 4\pi}$$

(10 marks)

Question 10 A guy wire runs from the ground to a cell tower. The wire is attached to the cell tower 150 feet above the ground. The angle formed between the wire and the ground is 43° .

a) How long is the guy wire?

b) How far from the base of the tower is the guy wire anchored to the ground?

(10 marks)

Question 11 Using the value of the trigonometric function, $\sin t = \frac{4}{5}$, to evaluate the indicated functions.

a) $\sin(t - \pi)$

b) $\sin(\pi + t)$ (10 mark)